Qiskit Certification — Visualization Module

30 Hard Multiple-Choice Questions

Study Group Prep Deck

August 29, 2026

Exam Practice — Question 1

Question 1

When `plot_histogram` is called with `data = [counts1, counts2]`, `legend = [...]` and `color=['crimson','midnightblue']`, how are colors assigned to datasets?

A. Colors are assigned alphabetically by bitstring, ignoring list order

B. The *i*-th color in `color` is mapped to the *i*-th dataset, matching the order of `data` and `legend`

C. Matplotlib assigns colors automatically and the `color` kwarg is ignored for multi-dataset input

D. Colors cycle based on the numeric magnitude of the counts, not dataset order

Exam Practice — Question 2

Question 2

Which statement correctly lists *all* valid ways (shown in the notebook) to persist a `plot_histogram` figure to disk?

A. Only passing `filename=...` directly to `plot_histogram`

B. Only calling `.savefig()` on the object returned by `plot_histogram`

C. Both: pass `filename` as a kwarg to `plot_histogram`, or capture its return value (a Matplotlib `Figure`) and call `.savefig()` on it

D. `plot_histogram` cannot save to disk; you must use `circuit_drawer` instead

Exam Practice — Question 3

Question 3

What is the core functional difference between `plot_histogram` and `plot_distribution`?

A. `plot_distribution` normalizes raw counts into a probability distribution before plotting; `plot_histogram` plots the raw counts as-is

B. `plot_distribution` only accepts `Statevector` objects, never counts dictionaries

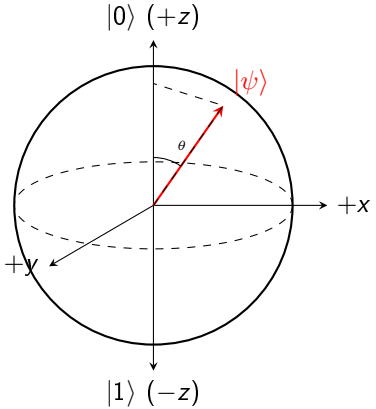
C. `plot_histogram` requires a live backend connection while `plot_distribution` does not

D. They are aliases of the same underlying function with no behavioral difference

Exam Practice — Question 4

Question 4

Given `plot_bloch_vector([1/np.sqrt(2), 1/np.sqrt(3), 1], coord_type='spherical')`, what do the three list elements represent, in order?



Exam Practice — Question 5

Question 5

If `coord_type` is *not* specified when calling `plot_bloch_vector`, what is assumed?

A. `spherical`
B. `cartesian`
C. `polar`
D. `cylindrical`

Exam Practice — Question 6

Question 6

For `qc.h(0); qc.x(1)` followed by `plot_bloch_multivector(Statevector(qc))`, what will the two Bloch spheres show?

A. A single combined 4-dimensional sphere, since the qubits interact

B. Two separate points, each on the *surface* of its own sphere, because the qubits are unentangled (the state is a simple product state)

C. Both points sitting exactly at the center of their spheres because the state is mixed

D. An error, since `plot_bloch_multivector` only supports single-qubit states

Exam Practice — Question 7

Question 7

For the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, why does `plot_bloch_multivector` show *no visible vector* (a point at the exact center) for each qubit?

A. `plot_bloch_multivector` cannot render entangled states at all

B. Each qubit's reduced density matrix is $\rho_i = I/2$, giving $\langle X\rangle = \langle Y\rangle = \langle Z\rangle = 0$, i.e. the Bloch vector $(0, 0, 0)$

C. The global phase of the Bell state cancels all vector components during plotting

D. `Statevector` objects default to the zero vector unless the circuit is measured first

Exam Practice — Question 8

Question 8

What does the `reverse_bits=True` argument do in `plot_bloch_multivector` (or `plot_state_city`)?

A. Flips the sign of every measured Pauli expectation value

B. Reverses the qubit display order (e.g. draws q_1 before q_0), inverting Qiskit's default little-endian ordering in the plot

C. Silently converts the input `Statevector` into a `DensityMatrix`

D. Mirrors each Bloch vector to the antipodal point on the sphere

Exam Practice — Question 9

Question 9

You call `plot_state_city(state, color=['red','green','blue'])` (three colors). What happens, and why?

A. `TypeError`, because `color` must be a plain string, not a list

B. `ValueError`, because `color` must be a tuple/list of exactly two entries (styling the real and imaginary bar sets)

C. Nothing — the extra color is silently ignored and the plot renders normally

D. `VisualizationError`, because colors must be given as hex codes only

Exam Practice — Question 10

Question 10

In `plot_state_city(state, alpha=0.6)`, what does the `alpha` parameter control?

A. The 3D camera rotation angle of the plot

B. The width of each bar in the city plot

C. The transparency of the bars, where 0 is fully transparent and 1 is fully opaque

D. The aspect ratio of the generated figure

Exam Practice — Question 11

Question 11

In a Hinton diagram (`plot_state_hinton`), what do square color and size encode?

A. White square = negative value, black square = positive value; size unrelated to magnitude

B. White square = positive value, black square = negative value; square size is proportional to the magnitude of that matrix entry

C. Color encodes the relative phase of each amplitude; size encodes qubit index

D. Square color encodes which qubit register the entry belongs to

Exam Practice — Question 12

Question 12

Given $\rho = \frac{1}{2^n} \sum_\sigma \mathrm{Tr}(\sigma\rho)\,\sigma$ over $\sigma \in \{I, X, Y, Z\}^{\otimes n}$, what exactly does each bar in `plot_state_paulivec` represent?

A. The eigenvalues of the density matrix ρ

B. The coefficient $\mathrm{Tr}(\sigma\rho)$ — the expectation value of each n-qubit Pauli string operator in the decomposition of ρ

C. The raw measurement probability of each computational basis state

D. Only the off-diagonal coherence terms of ρ, excluding diagonal probabilities

Exam Practice — Question 13

Question 13

If you pass a `color` list to `plot_state_paulivec` that has *fewer* entries than the number of bars, what happens?

A. A `ValueError` is raised immediately

B. The provided colors are cycled/alternated in sequence across the remaining bars

C. All remaining bars default to plain black

D. All remaining bars are rendered fully transparent

Exam Practice — Question 14

Question 14

In `plot_state_qsphere`, what do the *size* and *color* of each point represent, respectively?

A. Size encodes phase; color encodes probability

B. Size is proportional to the probability of that basis-state term; color encodes its phase

C. Size encodes the qubit count; color encodes amplitude magnitude only

D. Size and color both encode probability, redundantly

Exam Practice — Question 15

Question 15

What do `show_state_phases=True` and `use_degrees=True` do together in `plot_state_qsphere`?

A. `show_state_phases` hides phase info entirely; `use_degrees` converts probabilities into percentages

B. `show_state_phases` labels the phase value of each basis-state term directly on the qsphere; `use_degrees` displays that phase in degrees instead of the default radians

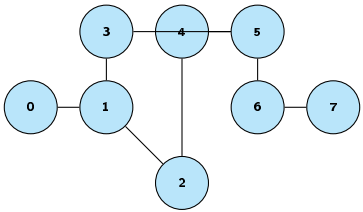
C. Both parameters only take effect for single-qubit states and are ignored otherwise

D. `use_degrees` rotates the entire 3D view by the number of degrees given

Exam Practice — Question 16

Question 16

What is the key structural difference between `plot_gate_map` and `plot_coupling_map`?



A. `plot_gate_map` needs only manually specified coordinates and a coupling list; `plot_coupling_map` always needs a live backend object

B. `plot_gate_map` visualizes the connectivity and native gates of an actual backend object; `plot_coupling_map` lets you plot an *arbitrary* coupling map from raw `num_qubits`, coordinates, and edge list, with no backend required

C. They are functionally identical wrappers around the same rendering code

D. `plot_coupling_map` only works for backends with a nonzero error map

Exam Practice — Question 17

Question 17

What information does `plot_error_map(backend)` display that `plot_gate_map(backend)` does not?

A. Only the physical qubit layout, with no numeric data at all

B. The gate and measurement error rates of the backend, overlaid on the connectivity graph

C. Only readout error, deliberately excluding two-qubit gate error

D. It requires a noise-free simulator and cannot be used on any noisy backend model

Exam Practice — Question 18

Question 18

To use `plot_circuit_layout` correctly, what must be supplied, and what does the resulting plot show?

A. Only the original (untranspiled) circuit; it shows the DAG of that circuit

B. A circuit that has already been `transpiled` for a target backend, plus that backend object; it shows how the circuit's virtual qubits were mapped onto the backend's physical qubits

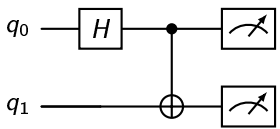
C. Only the backend object; the circuit argument is ignored

D. The backend's error map overlaid directly on top of the circuit diagram

Exam Practice — Question 19

Question 19

Regarding `circuit_drawer(qc, output="mpl", style={...})`, which statement is correct?



A. `output="mpl"` is the only rendering mode `circuit_drawer` supports; text output does not exist

B. The `output` argument selects the rendering backend (e.g. `text`, `mpl`, `latex`), and the `style` dict (e.g. `backgroundcolor`) customizes the appearance only when using the matplotlib renderer

C. An `mpl` output cannot be styled in any way

D. `circuit_drawer` always requires a backend object to run, even for plain drawing

Exam Practice — Question 20

Question 20

What is the fundamental purpose of `dag_drawer`, and how does it differ from `circuit_drawer`?

A. It shows gate timing across qubits like a Gantt/timeline chart

B. It visualizes *operation dependencies* as a Directed Acyclic Graph, a structural view distinct from `circuit_drawer`'s temporal, qubit-line view of the same circuit

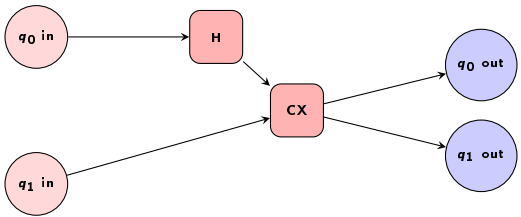
C. It is simply an alias for `circuit_drawer(qc, output='dag')`

D. It can only be used after a circuit has been transpiled

Exam Practice — Question 21

Question 21

In `dag_drawer(dag, style={"inputnodecolor":..., "outputnodecolor":..., "opnodecolor":...})`, what do these three style keys control?

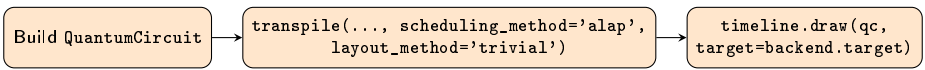


A. The colors of only measurement-type operation nodes
B. The colors of the DAG's input nodes, output nodes, and operation nodes respectively — the three structural node types in a `DAGCircuit`
C. The overall background color of the whole rendered figure
D. Which qubit vs. classical-bit wires are colored differently

Exam Practice — Question 22

Question 22

Before calling `qiskit.visualization.timeline.draw(qc, target=backend.target)`, what must first be done to qc?



A. Nothing — `draw()` performs scheduling automatically on any circuit

B. qc must be transpiled with an explicit scheduling method (e.g. `scheduling_method='alap'`) and a layout method against the target backend, so it carries concrete instruction timing

C. qc must first be converted into a `DAGCircuit` via `circuit_to_dag`

D. qc must contain only measurement instructions with no gates

Exam Practice — Question 23

Question 23

What distinguishes the `IQXSimple` and `IQXDebugging` style presets used with the timeline drawer?

A. They are backend-performance flags with no visual effect on the rendered output

B. They are alternate visual style presets offering different levels of instruction/timing detail in the rendered timeline

C. `IQXDebugging` only functions when paired with a statevector simulator backend

D. `IQXSimple` removes the qubit wires from the diagram entirely

Exam Practice — Question 24

Question 24

As documented in the notebook, what is the status of `qiskit.visualization.transition_visualization.visualize_transition()`?

A. It is actively maintained and recommended for all single-qubit state-transition animations

B. It is deprecated as of Qiskit 1.2.0 and scheduled for removal in the 2.0 release

C. It was moved into the `qiskit_aer` package and renamed

D. It was renamed to `plot_state_transition` starting in Qiskit 1.2.0

Exam Practice — Question 25

Question 25

What dimensionality constraint does `array_to_latex` place on its input numpy array?

A. It accepts arrays of any dimensionality (1D, 2D, 3D, ...)

B. It only accepts 1-dimensional or 2-dimensional arrays

C. It only accepts square 2D matrices, never vectors

D. It only accepts arrays of complex dtype; real-valued arrays raise an error

Exam Practice — Question 26

Question 26

In a density-matrix "city" plot (`plot_state_city`), what do the off-diagonal bars represent?

A. Measurement probabilities of the corresponding basis states

B. Quantum coherence between pairs of basis states

C. Labels identifying which physical qubit each bar belongs to

D. Two-qubit gate error rates from the backend's calibration data

Exam Practice — Question 27

Question 27

What do *strong* off-diagonal terms versus *no* off-diagonal terms in a density-matrix city plot indicate, respectively, about the underlying state?

A. Strong off-diagonal terms indicate a classical mixture; zero off-diagonal terms indicate a pure superposition

B. Strong off-diagonal terms indicate quantum superposition/coherence; zero off-diagonal terms indicate a classical mixture (no coherence between basis states)

C. Both cases are physically indistinguishable — the plot cannot differentiate them

D. Off-diagonal terms are always exactly zero for any pure state, regardless of superposition

Exam Practice — Question 28

Question 28

Physically, what does taking the partial trace $\rho_i = \mathrm{Tr}_{\bar{i}}(\rho)$ accomplish before plotting a per-qubit Bloch vector in `plot_bloch_multivector`?

A. It collapses the full multi-qubit state into a single classical bit value

B. It extracts qubit i's local, reduced single-qubit description by discarding (averaging over) its correlations with every other qubit, which is what enables one independent Bloch vector per qubit

C. It computes only the global phase of the state, discarding all amplitude information

D. It converts a `DensityMatrix` object back into a pure `Statevector`

Exam Practice — Question 29

Question 29

For a circuit `h(0); h(1); t(1); s(0); cx(0,1)`, plotting `plot_bloch_multivector(DensityMatrix(qc), reverse_bits=True)` shows both reduced states strictly *inside* the sphere (not on its surface). What does this indicate?

A. A bug in the plotting library that misplaces pure states

B. The reduced single-qubit states are mixed — because the two-qubit `cx` entangles the qubits — so each Bloch vector has length < 1 and lands inside the sphere

C. The overall two-qubit state failed to normalize correctly

D. `reverse_bits=True` discarded amplitude information, artificially shrinking the vectors

Exam Practice — Question 30

Question 30

Which single option correctly pairs *all four* of the following use-cases with the correct `qiskit.visualization` function: (i) measurement outcome counts, (ii) hardware error rates, (iii) operation-dependency structure, (iv) scheduled instruction timing?

A. (i) `plot_distribution`, (ii) `plot_gate_map`, (iii) `circuit_drawer`, (iv) `dag_drawer`

B. (i) `plot_histogram`, (ii) `plot_error_map`, (iii) `dag_drawer`, (iv) `timeline_drawer`

C. (i) `plot_state_city`, (ii) `plot_coupling_map`, (iii) `pass_manager_drawer`, (iv) `plot_circuit_layout`

D. (i) `array_to_latex`, (ii) `plot_bloch_vector`, (iii) `plot_state_hinton`, (iv) `plot_state_qsphere`